

PLANNING, SECURITY, AND YOUR FIRST REPOSITORY ENTRY

Submission Form

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Course/Section: BSIT - 80107	Date: September 2, 2026

Chosen App or Organization (semester project): Student Information System (SIS)

Part 1. Planning Question Evaluation

New request being evaluated (one sentence): Add a centralized online student portal where students can view grades, class schedules, enrollment information, and school announcements.

1. Does this solve a real problem? Yes. Students currently need to contact different offices or use separate systems to access academic information. A centralized portal would make information easier and faster to access.

2. Do we already have something that can do this? Some existing school systems may already provide individual functions such as grades or schedules, but they may not provide all required information in one integrated portal. The proposed system would improve integration and accessibility.

3. Does this support the organization's actual goals? Yes. It supports the organization's goals of improving student services, reducing manual administrative work, providing timely information, and improving digital access to school services.

4. Is it secure and cost-effective? Yes, if implemented with proper authentication, role-based access, encryption, backups, and regular security updates. Using existing school infrastructure and integrating current systems can also reduce development and maintenance costs.

Where would this rank against other approved work? (value, risk, dependency, or effort, reasoned in words, not calculated)

This would rank as a high-value, medium-priority project. It provides direct benefits to many students and staff and improves existing processes. However, it depends on the availability and integration of existing student, enrollment, and grading systems. Security and data privacy must also be considered because the system handles sensitive student information.

Part 2. Security Pass Across the Four Domains

Domain	Security Concern	Matching Control
Business	Unauthorized staff or students may access information they are not supposed to see.	Role-Based Access Control (RBAC) — only authorized users can access information based on their assigned role.
Data	Student grades, personal information, and enrollment records could be exposed or modified without authorization.	Data encryption and access controls — protect student information and allow only authorized users to view or modify records.
Application	Attackers could exploit vulnerabilities such as weak passwords, unauthorized requests, or improper input validation.	Strong authentication, input validation, and regular security testing — prevent unauthorized access and reduce application vulnerabilities.
Technology	Servers, databases, or network equipment could be compromised or become unavailable.	Firewalls, regular security updates, system monitoring, and backups — protect infrastructure and help maintain system availability.

One weak-link risk (how a weak point in one part could put a stronger part at risk): A weak application login system could allow an attacker to gain access to the student portal. Even if the database has strong security controls, compromised user credentials could allow unauthorized users to retrieve or modify sensitive student information. This shows that a weakness in the application layer can put the data and technology layers at risk.

Part 3. Your First Repository Entry

Component being documented (application, data store, or technology platform): Student Information Database

Field	Your Entry
What it does	Stores and manages student information, including personal details, enrollment records, class schedules, grades, and academic records.
Which department or user group uses it	Students, teachers, registrar staff, and authorized academic administrators.
What it connects to	Student Profile, Enrollment, Class Schedule, Grades, and User Account entities. These entities store and retrieve the information needed by the Student Information Database.
Who is responsible for it	The IT Department/System Administrator manages the database, while the Registrar and authorized school personnel manage and update student records.
Whether it is still needed, and why	Yes. It is needed because it provides a centralized storage system for student records and supports enrollment, scheduling, grading, and academic record management.